

Identification of IFITM3 as a diagnostic biomarker of systemic lupus erythematosus and its association with disease activity based on multi-omics and experimental verification

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Abstract

Background: Systemic lupus erythematosus is a clinically heterogeneous autoimmune disease that lacks reliable diagnostic biomarkers. In our study, we aimed to identify a novel biomarker for the diagnosis and disease activity monitoring of SLE. **Methods:** Bulk RNA and scRNA-seq datasets were obtained from the Gene Expression Omnibus database. In this study, differential analysis, cell-cell communication algorithm, functional enrichment analysis, human protein map database analysis, protein-protein interaction analysis and immune cell infiltration analysis were utilized to identify the hub genes between SLE and healthy groups. Furthermore, clinical data from 68 SLE patients and 31 healthy controls were collected for verification. Changes in IFITM3 levels were confirmed through quantitative real-time polymerase chain reaction, western blotting, and flow cytometry analyses.

Result: Bioinformatic analyses showed that IFITM3 expression was significantly upregulated in peripheral monocytes from patients with SLE. IFITM3 mRNA levels showed a significant diagnostic value for SLE, with an AUC value of 87.14%. IFITM3 expression was associated with the systemic lupus erythematosus disease activity index, as well as C3, C4, and IgG levels. The results of Chi-square test showed that those in the IFITM3-positive group had a higher percentage of several clinical manifestations such as thrombocytopenia, leukopenia, low complement, and fever.

Conclusions: These findings indicated an obviously increased expression of IFITM3 in peripheral blood monocytes of patients with SLE and verified IFITM3 as a promising diagnostic marker for SLE and associated with disease activity.

Keywords

Systemic lupus erythematosus, IFITM3, monocytes, multi-omics, disease activity

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Key point

- Bioinformatic analyses and molecular experiments verified the elevated expression level of IFITM3 in peripheral monocytes of SLE patients compared with HC individuals.
- The expression of IFITM3 was highly associated with SLE disease activity and clinical features.

Introduction

Systemic lupus erythematosus (SLE) is a multisystem autoimmune disease characterized by abnormal immune system activity that affects multiple organs and is often associated with a poor prognosis.^{1–3} The clinical manifestations of SLE are complex and varied, which complicates both diagnosis and treatment.⁴ However, the current SLE-related biomarkers still lack sufficient sensitivity, specificity and predictive power for clinical application. Although ANA tests in serum are essentially universal for classifying patients with SLE, Pisetsky et al. reported that up to 30% of patients with SLE screened in clinical trials for new therapies are ANA-negative.⁵ Serum C3 and C4 are widely applied to assess the presence of biologically active immune complexes, but, because of the low specificity of C3 and C4 in diagnosing SLE, the reliability of C3 and C4 levels as biomarkers for SLE diagnosis would be limited.⁶ Anti-dsDNA antibodies are regarded as the important biomarkers which are associated with SLE disease activity and can predict the development of lupus nephritis.⁷ Anti-dsDNA antibodies have a high specificity (96%) but low diagnostic sensitivity (52%–70%) for SLE as the level of anti-dsDNA antibodies would fluctuate over time.⁸ Consequently, it is necessary to discover novel biomarkers for SLE diagnosis and disease activity evaluation.

Type I interferon (IFN-I), particularly IFN- α , has been demonstrated to have a crucial role in the pathogenesis of SLE,⁹ which is largely produced by plasmacytoid dendritic cell. Once produced, IFN-I interacts with the interferon- α/β receptor (IFNAR, including subunits IFNAR1 and IFNAR2) expressed in most nucleated cells, and ultimately initiates the transcription of interferon-stimulated genes (ISGs) to generate IFN-I gene signature via JAK/STAT pathway,¹⁰ which can cause aberrant autoimmune responses and chronic inflammation. Moreover, genome wide association studies emphasized the contribution to SLE susceptibility of genes encoding molecules involved in nucleic acid degradation and sensing as well as IFN-I signaling pathways.¹¹ Zhao et al. found that DNA hypomethylation of two CpG sites of IFI44 L promoter could cause the elevated level of type-I IFN in SLE patient, which might be helpful in the diagnosis of SLE,¹² but this work did not link the levels of IFI44 L methylation with SLE disease activity. In addition, Shen et al. reported that four ISGs including IFI44,

USP18, RSAD2, and NRIR had good diagnostic efficacy for identifying SLE patients according to traditional bulk RNA-seq analyses and qRT PCR verification.¹³

The traditional bulk RNA-seq method reflects average gene expression but lacks single-cell resolution.¹⁴ Separating distinct cell subsets for investigating the activities of IFN signature could be more valuable for enriching pathogeny and developing targeted therapy for SLE patients. Recently, single-cell RNA-seq (scRNA-seq) as an innovative technology has been applied to study the transcriptome of different cell types, which allows expression profile of each cell type to be get rapidly.¹⁵ There are many differences in the disturbed cell subsets and proportions between SLE and healthy controls (HCs),^{16,17} so the identification of changed immune cell types in SLE helps us to understand the cellular results of the disease and establish the appropriate diagnostic or treatment strategies.

Our study aimed to identify the molecular biomarkers for SLE diagnosis through the integration of scRNA-seq and bulk RNA-seq data from PBMC samples. We identified 8 cell types from the scRNA-seq datasets and revealed alterations in several immune cell proportions between HCs and patients with SLE. In addition, we uncovered the differentially expressed genes (DEGs) between SLE and HC group in different PBMC cell subsets and performed GO and functional enrichment analyses to determine the functional signaling pathways of DEGs. Moreover, bulk RNA-seq data further aids in determining immune infiltration and hub genes, associated the predicting function of potential biomarkers with clinical features via conducting logistic analysis. Finally, qRT-PCR, Western blotting, and flow cytometry were conducted to validate the bioinformatic results.

Material and methods

Data collection

We downloaded scRNA-seq data of PBMC samples from Gene Expression Omnibus (GEO) database. scRNA-seq data included 5 SLE patients and 5 HC samples were obtained from GSE142016, GSE162577, and GSE185857 dataset. In addition, we downloaded bulk RNA-seq of PMBCs of SLE and HC samples from GEO database, including GSE160080 and GSE139350. We used the GSE65391 dataset to explore the traditional bulk RNA-seq links between clinical features and selected potential biomarkers for SLE. GSE122459 was used to perform weighted Gene Co-expression Network Analysis (WGCNA).

Human single-cell RNA sequencing analysis

Processing and preliminary analysis of scRNA-seq data from SLE and HC group. After downloading scRNA-seq data from

GEO database, at first, cells were excluded based on the “Seurat” scRNA-seq analysis workflow criteria¹⁸: (1) The number of measured genes in each cell was less than 300. (2) The mitochondrial contamination was more than 10%. The filtered cells were selected for creating a Seurat object and normalized using the “NormalizeData” function, 3000 highly variable genes were screened employing the “FindVariableFeatures” function. Then, the canonical correlation analysis (CCA) method including the “FindIntegrationAnchors” and “IntegrateData” functions were applied to remove batch effects after integrating three scRNA-seq datasets. We obtained a total of 45,927 cells after removing the batch effects. We then applied principal component analysis (PCA) for dimensionality reduction, selecting the top 50 principal components as input for the uniform manifold approximation and projection (UMAP) algorithm, and visualized the first two UMAP dimensions at a clustering resolution of 0.5. Cell clusters were determined using the “FindNeighbors” and “FindClusters” function and categorized into 21 different subclusters. Next, “SingleR” package was exploited to annotate each cell cluster to recognize different subtypes.¹⁹ “FindMarkers” function was used to identify DEGs ($|\log_{2}FC| > .25$) between SLE and HC group.

Cell-cell communication analyses. R package “CellChat” was adopted to identify and visualize cell–cell interaction of distinct cell types under SLE and HC conditions.²⁰ We followed the official workflow to input the normalized count from merged Seurat object into “CellChat” and started the standard preprocessing steps. Then functions “identify Over Expressed Genes”, “identify Over Expressed Interactions” and “project Data” with default parameters were applied to obtain the processed data. Our work selected the “Cell-Cell Contact” interaction from CellChatDB as referencing database and used the precompiled human protein–protein interactions (PPIs) as a priori network information. The important functions including “compute CommunProb”, “compute CommunProbPathway” and “aggregate Net” were applied setting default parameters. Then we visualized the count of interactions and showed the aggregated cell-cell communication network and signaling sent from each cell type.

Human bulk RNA sequencing analysis

Differentially expressed genes identification. DEGs from RNA-seq were filtered by using “DESeq2” R package,²¹ the “limma” package was applied to normalize and analyze the DEGs from the microarray dataset. $|\log_{2}FC| > 2.0$ and false discovery rate (FDR) < 0.05 were set as the cut-off value.

Immune cell infiltration analysis. “CIBERSORT” algorithm is commonly used to perform immune cell infiltration

analysis.²² “CIBERSORT” R package was exploited to calculate the proportion of 22 immune cells in all samples from bulk RNA-seq.²² The results of immune infiltration proportion were visualized via the function of “barplot”.

Identification of the hub genes among bulk RNA-seq and scRNA-seq datasets. The filtered DEGs were inputted into “Venny” online tool (<https://bioinfogp.cnb.csic.es/tools/venny/index.html>) for getting the overlapping genes among bulk RNA seq and scRNA seq datasets. Then, PPI network of hub genes was constructed via applying STRING (<https://string-db.org/>).²³ Then, volcano plots were used to display the distribution of the overlapping genes in two bulk RNA seq datasets. In addition, the expression of IFITM3 in immune cells at transcriptional and single-cell level was detected using HPA database.²⁴

Weighted gene co-expression network analysis of IFITM3-low and IFITM3-high groups. “WGCNA” method was performed to identify gene modules related to IFITM3 expression level. Co-expression gene modules were determined using the “WGCNA” R package.²⁵ At first, we constructed a scatter plot of the fit index to determine an optimal soft power and the most suitable weighting coefficient β to establish a weighted adjacency matrix and then transformed it into a topological overlap matrix (TOM). Modules acquired using the TOM dissimilarity measure (1-TOM) according to the hierarchical clustering tree algorithm. Then, we calculated the module eigengene (ME) of every module and the correlation between ME and IFITM3 expression level.

Functional enrichment analyses of significant IFITM3-related gene modules. “ClusterProfiler” R package was applied to conduct Gene Ontology (GO) enrichment analysis including biological process (BP), cellular component (CC), molecular function (MF) for significant IFITM3-related gene modules,²⁶ the threshold was set at $p < .05$, and the species was limited as Homo sapiens. Kyoto Encyclopedia of Genes and Genomes (KEGG) is a widely used database for systematic analyses of gene functions, which cluster functional gene sets into different molecular mechanisms and biological processes. We imported the gene sets into “ClueGO” (a plug-in of “Cytoscape” software)²⁷ for the KEGG and REACTOME pathway enrichment analyses, $p < .05$ was set as a threshold.

Validation of mRNA expression of IFITM3 in clinical samples

We collected PBMC samples of 68 SLE patients and 31 HCs from First Affiliated Hospital of Xiamen University. The study was reviewed and approved by the Ethics

Committee of the First Affiliated Hospital of Xiamen University (KY-2019-023).

qRT-PCR was used to quantify the expression of IFITM3. Total RNA was extracted from PBMCs using TRIzol Reagent (ambion, USA). Reverse Transcription System (Roche, Germany) was applied for perform reverse transcription PCR. PCR was performed using a LightCycler PCR cycler, qRT PCR with FastStart™ Universal SYBR® Green (ROX) (Roche, Germany) was performed. The cycle threshold (CT) data was determined, and the mean CT was determined from triplicate PCRs. Relative gene expression was calculated with the equation $2^{-\Delta\Delta CT}$. The following are primer sequences: IFITM3 (forward primer: TGCCTGGGC TTCATAGCATT; reserve primer: TCAGCACTGGGA TGACGATG); ACTIN (forward primer: CATGTACGT TGCTATCCAGGC; reserve primer: CTCCTTAATGTC ACGCACGAT).

Western blotting analysis

Total cellular proteins were extracted by RIPA (Solarbio, China) with cOmplete protease inhibitor (Roche, Germany) and phosphorylase inhibitor (Roche, Germany). BCA (Thermo, USA) reagent was utilized to evaluate protein concentration. The proteins were separated by 15% SDS polyacrylamide gel electrophoresis (SDS-PAGE) and then transferred onto the PVDF membrane (Millipore, Temecula, CA, USA). The membranes containing proteins were blocked with 5% not-fat milk at room temperature for 1h and the membrane was incubated with diluted IFITM3 polyclonal antibody (Proteintech, China) overnight on the shaker at 4°C. After washing thrice with Tris-buffered Saline with Tween, the membrane-containing proteins were incubated on the shaker with the secondary antibody for approximately 1 h at room temperature. After cleaning the membrane, we used the ChemiDoc Imaging Systems (Bio-Rad, USA) to detect targeted protein bands.

Validation of IFITM3 with flow cytometric analysis

PBMC samples from 5 healthy control, 5 SLE patients were obtained from First Affiliated Hospital of Xiamen University. PBMC samples were incubated with conjugated antibiotics including CD4-APC, CD8-PE-Cy™5.5, CD56-PE-Cy™7, CD19-BV421, CD14-PE, IFITM3-FITC (Biollegend, USA). For IFITM3 we used a two-step staining. After the primary surface antigen staining, cells were fixed with fixation buffer (BD, USA) for 20 min. Permeabilization buffer (BD, USA) was then diluted at a ratio of 1:10 with distilled water and cells were then incubated with the diluted permeabilization buffer for 10 min. Then, cells were incubated with the antibodies for intracellular antigen staining, diluted in permeabilization buffer for 15 min. Data was acquired on a CytoFLEX Beckman

Coulter flow cytometer (Beckman Coulter, USA) and analyzed with FlowJo V10. The expression of IFITM3 was identified by mean fluorescence intensity (MFI).

Statistical analysis

Statistical analyses were conducted using R software (v 4.1.3), results were displayed and visualized using the R packages. Differences between groups were analyzed by *t* test and Chi-squared test. The correlation studies were analyzed by Spearman Correlation coefficient. A *p* value <.05 was considered statistically significant.

Results

Single cell clustering, cell subsets annotation and identification of marker genes

10× scRNA-seq data of 5 SLE and 5 HC PBMC samples were obtained from the GSE142016, GSE162577 and GSE185857 datasets, a total of 45,927 cells were identified after quality control. We implemented PCA analysis to reduce technical noise in the scRNA-seq dataset. The most appropriate number of PCs (50 PCs) was selected for downstream analyses via computing the standard deviation of all principal components. In addition, we applied UMAP (a nonlinear dimensionality reduction algorithm) to visualize the distribution of single-cell data, and 21 distinct clusters were classified after PCA and UMAP analyses (Figure 1(a)). Cell types were annotated using “SingleR” R package and UMAP algorithm was used to visualize the cell types after dimensionality reduction. These 21 clusters were categorized into 8 cell types: CD4⁺ T cells, CD8⁺ T cells, Monocytes, NK cells, B cells, dendritic cells, platelets, and Neutrophils (Figure 1(b)). Then, we counted the frequency of these 8 cell subclusters and revealed a higher proportion of B cells and monocytes in SLE samples versus HC samples (Figure 1(c)).

Differential and cell-cell communication analyses

We screened the DEGs for each cell type between the SLE and HC group, and we identified 5921 DEGs. The top 5 significant DEGs of distinct cell types was shown in Figure 1(d). The heatmap showed the top 5 upregulated and downregulated DEGs of different SLE and HC PBMC samples (Figure 1(e)). In addition, we investigated the cell-cell communication network by computing communication probabilities. Compared with the HC group, there were more interactions between monocytes and B cells in the SLE group and fewer interactions between T cells and NK cells (Figure 1(f)).

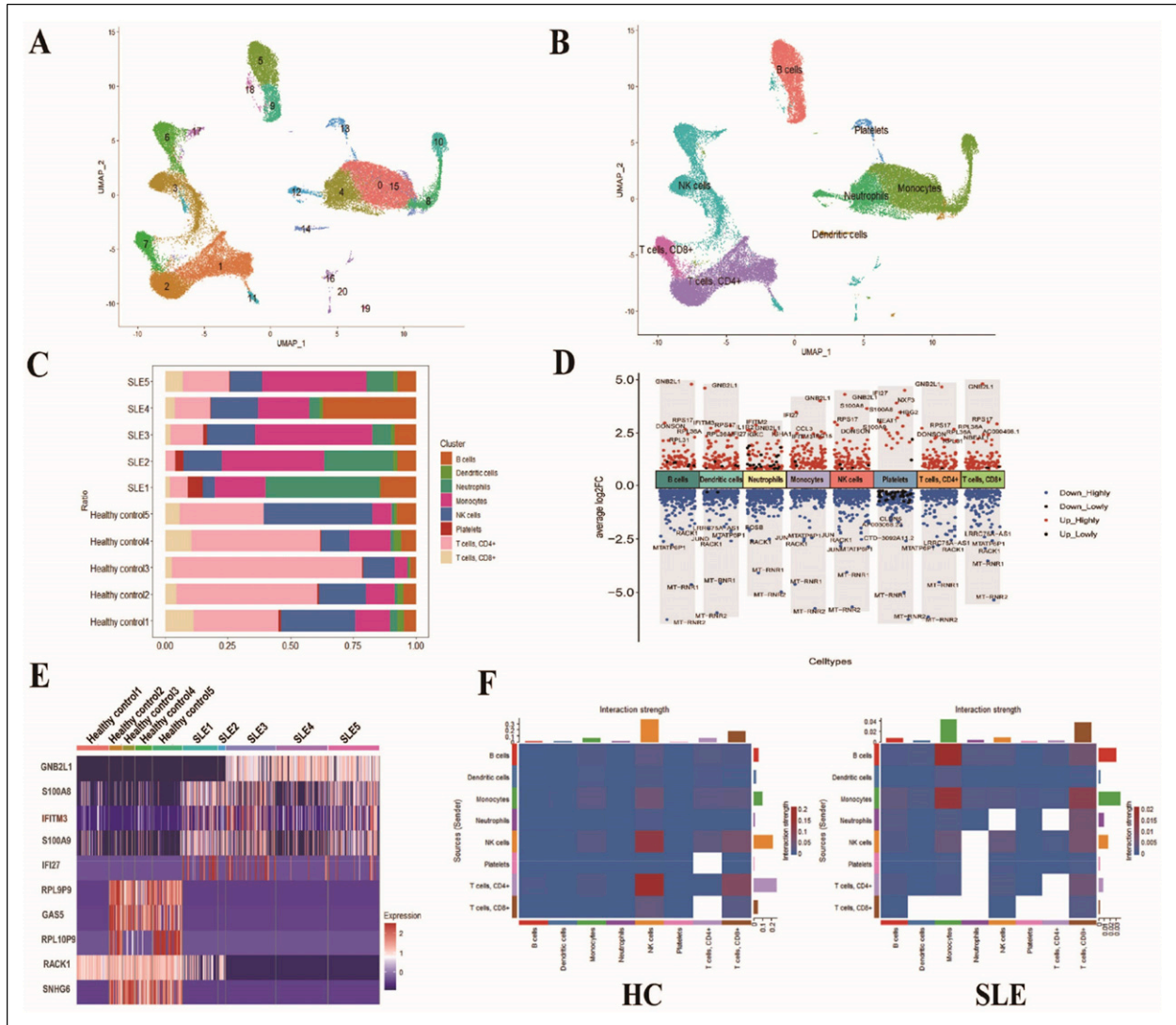


Figure 1. Single-cell clustering, annotation of cell subsets, and identification of marker genes. The UMAP visualization of 21 subclusters (a). The UMAP visualization of 8 cell types (b). Bar plots showed the proportion of different cell types in each sample (c). The top 5 DEGs of each PBMC cell type between SLE and HC groups (d). The heatmap showed the top 5 upregulated and downregulated DEGs of different SLE and HC PBMC samples (e). Cell-cell communication analysis between SLE and HC groups (f).

Bulk RNA-seq analysis and identification of the hub genes

Differential gene expression analysis of bulk RNA-seq datasets identified DEGs between the SLE and HC groups. We found 994 DEGs ($p < .05$ and $\log_2FC > 2$ or less than -2) in GSE160080 dataset and 622 DEGs ($p < .05$ and $\log_2FC > 2$ or less than -2) in GSE139350 dataset. Subsequently, we adopted the CIBERSORT algorithm to calculate the proportion of 22 immune cells in all samples. The results demonstrated that the infiltration levels of B cells and monocytes were increased in the SLE group compared with HC group (Figure 2(a)). We used Venn plot to identify the overlapping DEGs among two bulk RNA-seq datasets and merged scRNA-seq dataset, 20 overlapping genes were

screened for later analyses (Figure 2(b)). Then, we excluded the disconnected nodes in the network using “STRING” tool and visualized the PPI network of the hub DEGs, including IFIT3, IFITM3, RSAD2, S100A8, S100A12, S100A9, EPSTI1, IFI27, and IFIT1 (Figure 2(c)). Finally, volcano plots were utilized to exhibit the key biomarkers in two bulk RNA-seq datasets (Figure 2(d)).

Associations between hub genes and systemic lupus erythematosus disease activity index of SLE

We adopted correlation analyses to investigate associations between hub genes and the SLEDAI score. The result

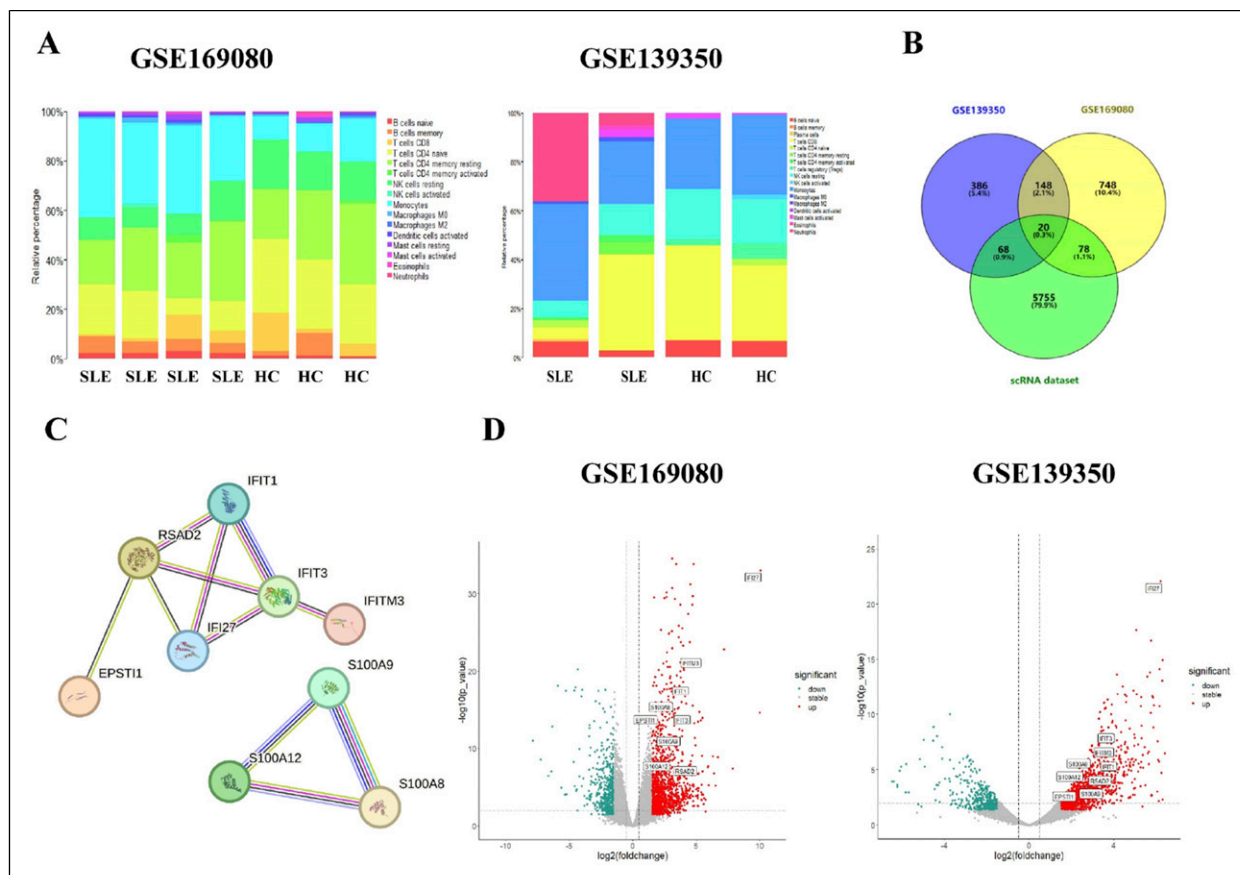


Figure 2. Bulk RNA-seq data analyses. Bar plots showed the proportion of different immune cell types in each sample (a). Venn plot identified the overlapping DEGs among two bulk RNA-seq datasets and scRNA-seq dataset (b). PPI network of the overlapping DEGs (c). Volcano plots exhibited the hub genes in two bulk RNA-seq datasets (d).

showed that the expression levels of IFITM3, RSAD2, S100A12, IFI27 and EPSTI1 were significantly positively correlated with the SLEDAI score (Figure 3). Among them, there is a strong correlation between IFITM3 and SLEDAI score, so IFITM3 may play an important role in diagnosing and predicting the disease activity of SLE. In addition, the increased expression of IFITM3 in the monocytes of patients with SLE was identified by scRNA-seq transcriptome analyses (Figure 4(a) and (b)). Finally, we used HPA database to investigate which immune cells express IFITM3 at transcriptional and single-cell level, and we found that IFITM3 was highly expressed in monocytes compared with other immune cells according to the HPA dataset, Schmiedel dataset, and flow sorted data. (Figure 4(c)).

Weighted co-expression network construction and identification of IFITM3-related gene clusters

We applied the Pearson's correlation coefficient to cluster the SLE samples dataset. After removing outliers, we drew a sample clustering tree (Supplemental Figure 1A). We then

obtained adjacency matrix and established the topological overlap matrix (Supplemental Figure 1B). Next, 36 modules were selected according to average hierarchical clustering and dynamic tree clipping (Supplemental Figure 1C). The p value of each module less than 0.05 was selected as the cluster highly related to the expression level of IFITM3 for subsequent analyses.

Functional enrichment analyses of IFITM3-related gene modules

Functional enrichment analyses including GO, KEGG, and Reactome were performed on the IFITM3-related gene sets to illustrate the specific biological functions and pathways. Results of GO analysis indicated that IFITM3-related genes might participate in the immune response-regulating signaling pathway, regulation of immune effector process, immune response-activating signaling pathway, regulation of innate immune response, positive regulation of cytokine production, activation of immune response (Supplemental Figure 2A). The pathway enrichment analyses showed that

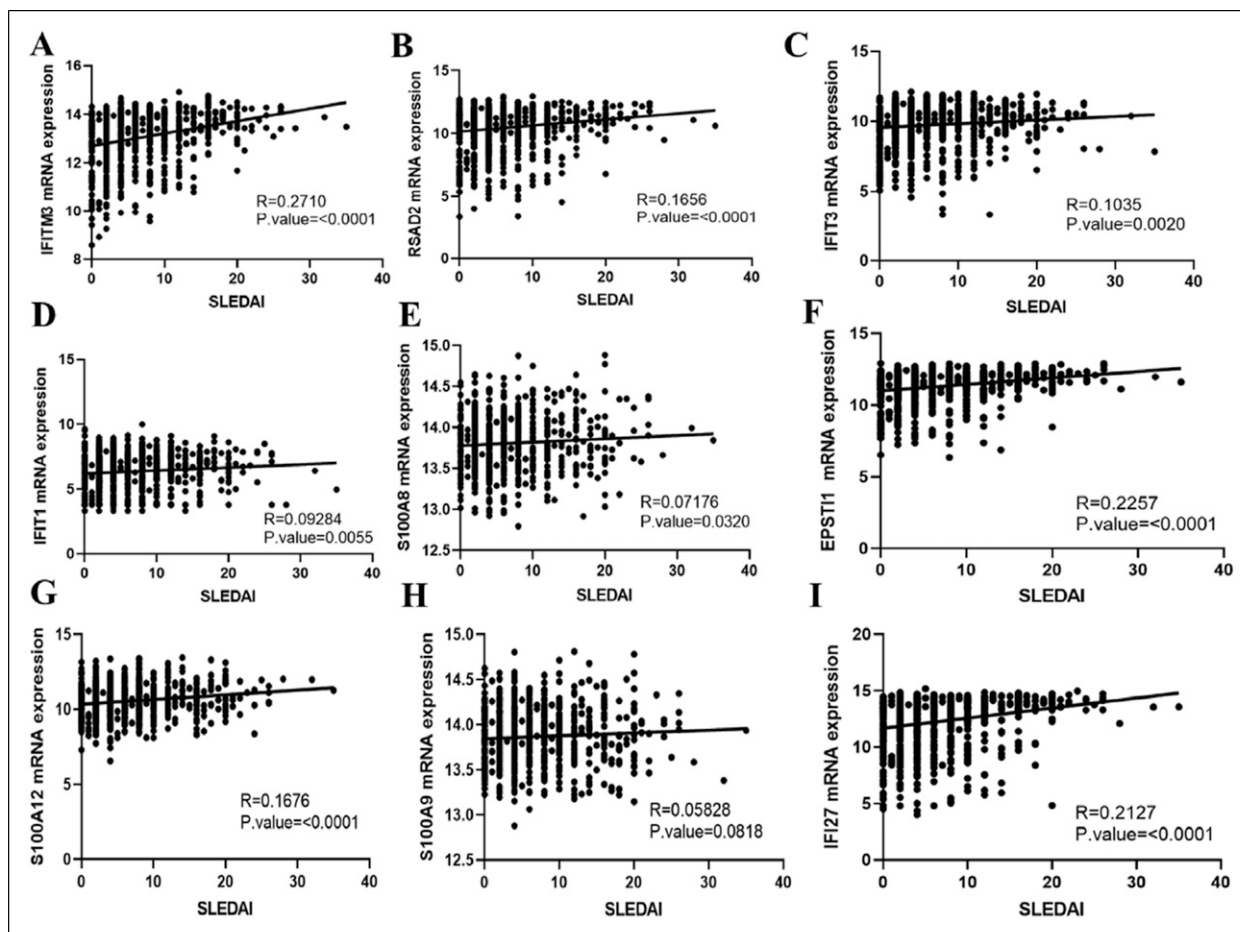


Figure 3. Correlation analyses between hub genes and SLEDAI score.

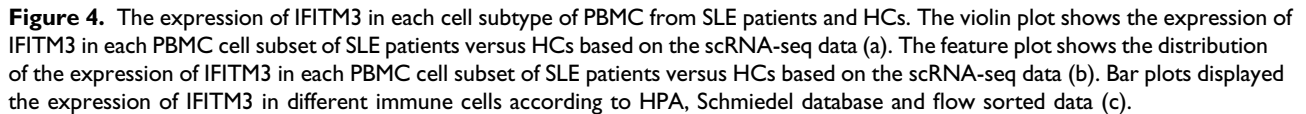
IFITM3-related genes might be involved in antigen processing-cross presentation, cytokine signaling in immune system, adaptive immune system, innate immune system, antigen processing: ubiquitination & proteasome degradation, class I MHC mediated antigen processing & presentation, signaling by the B Cell receptor and so on (Supplemental Figure 2B).

Validation of the expression of IFITM3 in clinical samples

PBMC samples of 68 patients with SLE and 31 HC were collected to detect the expression of IFITM3. The qRT PCR results showed that compared to the HC group, the mRNA expression levels of IFITM3 in the PBMCs of the SLE group were significantly increased (Figure 5(a)). Further, the protein level of IFITM3 was significantly increased in PBMC from patients with SLE (Figure 5(b)). It suggests that IFITM3 in monocytes might be involved in the development of SLE and serve as a key diagnostic biomarker.

Evaluation and verification of the diagnostic effect of IFITM3 on SLE

The Spearman correlation analysis was carried out to verify the correlation between IFITM3 and clinical features. The baseline characteristics of all samples analyzed in our study are exhibited in Table 1. The results of correlation analysis demonstrated that the expression of IFITM3 was significantly positively correlated with SLEDAI and IgG (Figure 5(c)), it was significantly negatively correlated with C3, C4 (Figure 5(d) and (e)). In addition, the expression of IFITM3 was significantly positively correlated with IgG (Figure 5(f)). We used the ROC curves to evaluate IFITM3 diagnostic efficiency. As shown in Figure 5(g), the AUC value of IFITM3 was 87.14%, with sensitivity of 72.46% and specificity of 93.75%. As shown in Figure 5(h), the ROC curve for IFITM3 demonstrates its potential as a biomarker to stratify patients with SLEDAI >4 (high disease activity) and SLEDAI ≤4 (low disease activity). The AUC value of IFITM3 was 88.96%, indicating excellent diagnostic performance. The sensitivity and specificity were 85% and 87.5%, respectively, further supporting the utility



We filtered the IFITM3-related gene modules and performed GO and pathway enrichment analyses to detect the functional signaling pathways about these gene modules. We found that IFITM3 could involve in B-cell receptor (BCR) signaling, SLE is featured as losing B cell tolerance and autoantibody production, specific pathways which negatively regulate B cell signaling have been reported to be

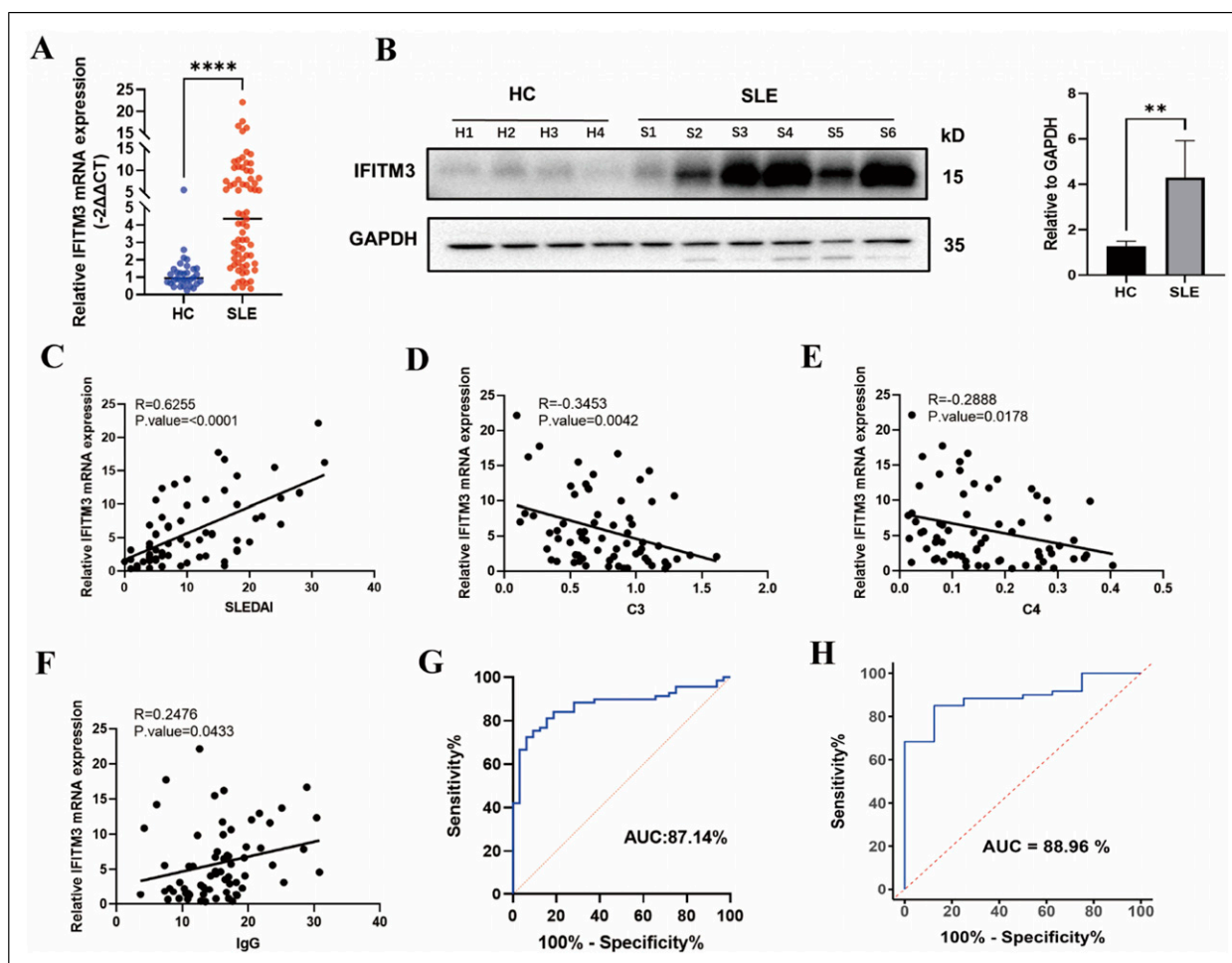


Figure 5. Validation of the expression of IFITM3 in clinical samples. qRT PCR results showed that the mRNA expression levels of IFITM3 in PBMCs of SLE patients compared with HCs (a). Western blotting results showed that the protein levels of IFITM3 in PBMC samples from SLE patients compared with HCs (b). The possible correlation between the expression of IFITM3 and SLEDAI (c). The possible correlation between the expression of IFITM3 and C3 (d). The possible correlation between the expression of IFITM3 and C4 (e). The possible correlation between the expression of IFITM3 and IgG (f). ROC curve demonstrating the potential of IFITM3 as a biomarker to stratify patients with SLEDAI >4 and SLEDAI ≤4 (h).t test and Spearman correlation analyses were used to analyze data for differences. Data are representative of 3 independent experiments. (** $p < .01$; ****, $p < .0001$).

impaired in some SLE patients.³¹ In normal resting B-cells, IFITM3 was minimally expressed and mainly localized in endosomes. However, BCR engagement would induce expression of IFITM3 and phosphorylation at Y20, resulting in accumulation at the cell surface.³² In addition, Jae Woong Lee team reported that IFITM3-dependent amplification of PI3K signaling would play a critical role in downstream of the BCR, it is important for the rapid expansion of B cells with high affinity to antigen.³³ Thus, we consider that IFITM3 may relate to B cell tolerance losing and autoantibody production.

In this study, we confirmed that IFITM3 expression is significantly upregulated in patients with SLE and is strongly correlated with disease activity and levels of serum

IgG. Based on these findings, IFITM3 may serve not only as a diagnostic biomarker but also as a therapeutic target. One study has shown that IFITM3 enhances BCR signaling, promoting B cell activation and proliferation, which in turn leads to autoantibody production.³³ Therefore, we hypothesize that inhibition of IFITM3 could potentially reduce BCR signaling activity, thereby decreasing autoantibody generation. Moreover, inhibiting IFITM3 has shown therapeutic potential in other immune-related diseases. For instance, IFITM3 deletion has been demonstrated to attenuate the antibody response after vaccination,³⁴ further supporting the role of IFITM3 in regulating antibody production. In the context of SLE, targeting IFITM3 with specific inhibitors or gene-silencing techniques may help

Table 1. Baseline characteristics of all subjects.

Characteristics	Patients with SLE (n = 68)	HCs (n = 31)	p-Value
Age (mean $\pm$ SD)-years	34.45 $\pm$ 13.36	32.13 $\pm$ 6.62	0.6369
Sex (%) - female	93.75%	88.89%	0.5347
Height-cm	161.7 $\pm$ 8.09	163.1 $\pm$ 7.49	0.6268
Weight-kg	56.24 $\pm$ 11.77	53.22 $\pm$ 9.148	0.4757
SLE disease duration (years), mean (SD)/median	5.92 (5.80)/5	/	/
Renal failure (%)	4.41%	/	/
Disease activity, mean SLEDAI	10.8 $\pm$ 7.95	/	/
SLEDAI $\leq$ 4 (%)	16 (25%)	/	/
SLEDAI >4 (%)	52 (75%)	/	/
Treatments (n)	Prednisone: n = 24 hydroxychloroquine: n = 30 mycophenolate: n = 6 rituximab: n = 3 Untreated: n = 20	/	/
Prednisone dose, mean (SD), mg/day	14.43 (9.35)	/	/
ANA positive (%)	88.09%	/	/
Anti-dsDNA antibody positive (%)	42.85%	/	/
Anti-Sm antibody positive (%)	26.19%	/	/
Anti-histone antibody positive (%)	38.09%	/	/
Anti-nucleosome antibody positive (%)	55.00%	/	/
Anti-SSA antibody positive (%)	45.23%	/	/
Anti-SSB antibody positive (%)	9.52%	/	/
Anti-Ro52 antibody positive (%)	40.48%	/	/

reduce inflammation and autoantibody levels, thus ameliorating disease symptoms and slowing disease progression. Preclinical studies using animal models of SLE will be critical to evaluating this therapeutic strategy.

SLE is a heterogeneous disease with a wide spectrum of clinical manifestations, making it crucial to identify subgroups of patients who may respond differently to specific treatments. There are emerging studies about a role of endogenous production of type I IFNs in the pathogenesis of neutropenia and lymphopenia in SLE,^{35,36} elevated serum levels of IFN- α have been detected in SLE and it is highly correlated with fever, skin rash and leucopenia.³⁶ Our study demonstrates that IFITM3 expression is significantly correlated with key clinical features of SLE, including fever, hematuria, proteinuria, low complement levels, leukopenia, and thrombocytopenia. Notably, IFITM3 expression is strongly associated with SLEDAI scores, allowing us to stratify patients based on disease activity. Patients with SLEDAI scores >4 (high disease activity) exhibit significantly higher levels of IFITM3 expression and more severe clinical features. In contrast, patients with SLEDAI scores $\leq$ 4 (low disease activity) show lower IFITM3 expression and milder clinical presentations. These findings suggest that IFITM3 could serve as a biomarker to define subgroups of SLE patients based on disease activity. Patients with high IFITM3 expression may require more aggressive therapeutic strategies due to the greater severity of their clinical

manifestations. Further validation in larger patient cohorts is necessary to confirm these findings and to explore the therapeutic potential of targeting IFITM3 in different SLE subgroups.

Monocytes play a particularly central role in the immune dysregulation observed in SLE.³⁷ Their involvement in antigen presentation, cytokine production, and inflammation is widely reported in the context of SLE.³⁸ IFITM3 upregulation in monocytes has been closely linked to their activation and inflammatory potential, which are pivotal processes in SLE pathophysiology. In addition, monocytes are one of the more abundant immune cell populations in PBMCs,³⁹ which made them more practical for experimental validation and data collection. But it is important to recognize that IFITM3 is also upregulated in other immune cell types, including neutrophils and dendritic cells.^{32,33,40} In particular, the enhanced expression of IFITM3 in dendritic cells may lead to enhanced presentation of self-antigens, thereby exacerbating autoreactive T cell responses and driving autoimmune inflammation in SLE.⁴¹ Neutrophils in SLE have been shown to produce neutrophil extracellular traps (NETs), which can release DNA and nuclear components that promote autoantibody production.⁴² IFITM3 upregulation may lead to increased NET formation. This may exacerbate autoimmunity by providing additional autoantigens for recognition by the immune system, leading to a vicious cycle of inflammation and

Table 2. Associations between IFITM3 expression and clinicopathological characteristics of patients with SLE ($n = 68$) (*, $p < .05$).

Clinicopathological characteristics	Number	IFITM3		χ^2	p -value
		Positive (42)	Negative (26)		
Arthritis				0.08303	0.7732
Positive	42	11	31		
Negative	26	6	20		
Thrombocytopenia				4.72	0.0298*
Positive	42	10	32		
Negative	26	1	25		
Leukopenia				5.868	0.0154*
Positive	42	14	28		
Negative	26	2	24		
Hematuria				3.392	0.0655
Positive	42	19	23		
Negative	26	6	20		
Proteinuria				2.873	0.0901
Positive	42	20	22		
Negative	26	7	19		
Skin lesions				2.102	0.1471
Positive	42	15	27		
Negative	26	5	21		
Low complement				5.283	0.0215*
Positive	42	31	11		
Negative	26	12	14		
Fever				4.756	0.0292*
Positive	42	19	23		
Negative	26	5	21		
Anti-dsDNA antibody				1.754	0.1854
Positive	42	18	24		
Negative	26	7	19		
Serositis				3.232	0.0722
Positive	42	8	34		
Negative	26	1	25		
Vasculitis				0.7599	0.3834
Positive	42	4	38		
Negative	26	1	25		
Oral ulcer				3.232	0.0722
Positive	42	8	34		
Negative	26	1	25		
Myositis				0.7177	0.3969
	42	4	38		
	26	1	25		
Alopecia				1.081	0.2985
	42	9	33		
	26	3	23		

Associations between IFITM3 expression and clinicopathological characteristics of patients with SLE ($n = 68$) cut off = mean of HC + 2 folds of SD (3.109).

tissue damage. Future studies should explore the specific functions of IFITM3 in these various cell types and their contribution to the overall immune dysregulation observed in SLE.

There are several limitations in this work, as excellent diagnostic biomarkers should be highly sensitive and

specific for a disease, IFITM3 as a potential diagnostic biomarker only has high specificity but low in sensitivity, so it should be combined with other ideal biomarkers with high sensitivity in SLE diagnosis. In addition, the cohort of SLE patients and HCs in our study was limited, we plan to include larger numbers of patients and HCs in

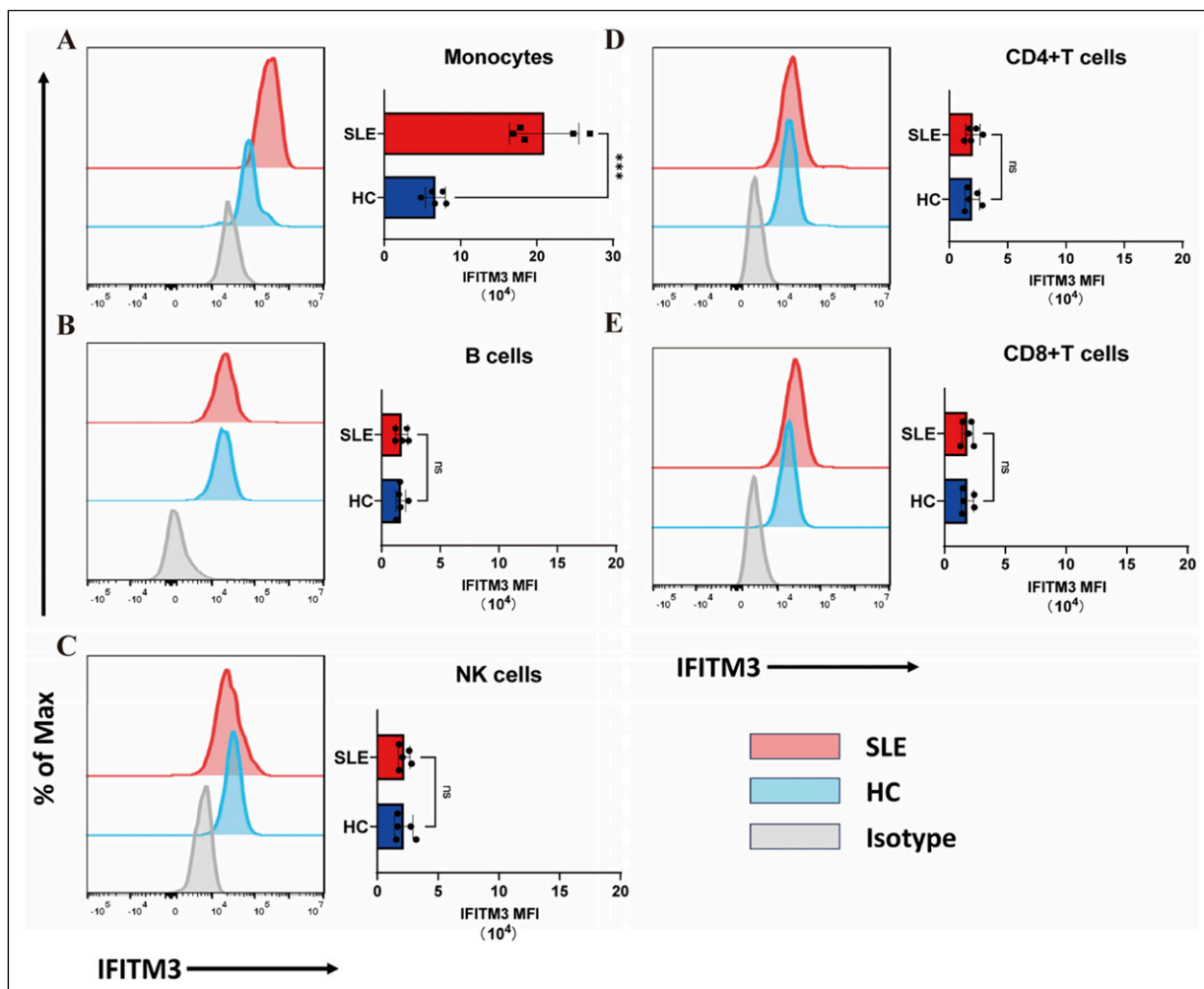


Figure 6. Flow cytometry analysis was applied to detect the expression of IFITM3 in different cell subsets of PBMC. Representative graph and MFI of IFITM3 expression on CD14⁺ monocytes (a). Representative graph and MFI of IFITM3 expression on CD19⁺ B cells (b). Representative graph and MFI of IFITM3 expression on CD56⁺ NK cells (c). Representative graph and MFI of IFITM3 expression on CD4⁺ T cells (d). Representative graph and MFI of IFITM3 expression on CD8⁺ T cells (e). *t* test analysis was used to analyze data for statistic differences ($n = 5$). Data are representative of 3 independent experiments. PBMC from different individuals were used for each independent experiment. (* $p < .05$; ns, not significant).

future studies to raise the accuracy and reliability of the results.

Conclusion

In conclusion, this work described the immune features of PMBCs under SLE and HC conditions, which highlighted the immune infiltration influences in the peripheral immune environment in SLE. In addition, we identified and verified the elevated expression level of IFITM3 in peripheral monocytes of SLE patients compared with HC individuals via bioinformatic analyses and molecular experiments. Furthermore, we revealed that the expression of IFITM3 was highly associated with SLE disease activity and clinical features, which inferred that IFITM3 had the

potential to become a novel biomarker for SLE diagnosis and disease activity evaluation. In summary, our work may contribute to advances in the diagnostic and disease activity evaluation of SLE patients.

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Authors' contributions

Yan Li and Jimin Zhang contributed equally. YL and JM contributed to write the manuscript and perform experiments. GS and

XW reviewed and edited the manuscript. XM contributed to clinical sample collection. All authors reviewed the manuscript.

Declaration of conflicting interests

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Ethical Statement

Ethical Approval

The study was reviewed and approved by the Ethics Committee of the First Affiliated Hospital of Xiamen University (KY-2019-023).

Informed Consent

All the participants have given approval for publication.

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Supplemental Material

Supplemental Material for this article is available online.

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Appendix

Abbreviations

SLE	Systemic lupus erythematosus
HC	Healthy control
qRT PCR	Quantitative real-time polymerase chain reaction
scRNA	Single-cell RNA
DEGs	Differentially expressed genes
GEO	Gene Expression Omnibus
CCA	Canonical correction analysis
PCA	Principal component analysis
UMAP	Uniform manifold approximation and projection
PPIs	Protein–protein interactions
FDR	False discovery rate
SLEDAI	Systemic lupus erythematosus disease activity index
CT	Cycle threshold
MFI	Mean fluorescence intensity
BP	Biological process
CC	Cellular component
MF	Molecular function
TLR	Toll-like receptor
IFITMs	Interferon-induced transmembrane proteins
IFNs	Interferons
JAK	Janus kinase
STAT	Signal transducer and activator of transcription
ISGs	IFN-stimulated genes
WGCNA	Weighted gene co-expression network analysis
GO	Gene ontology
KEGG	Kyoto encyclopedia of genes and genomes